Body Temperature Homeostasis



Thermoregulation

Thermoregulation

- Regulation of core body temperature at 36.8^{o}C
  - Temperature where chemical reactions in cells are stable and optimised.
- Homeostatic mechanisms do this.
- Balance between heat production and heat loss.
- If body temp too high:
  - Proteins denature
  - Nerve cells do not function correctly
  - Chemical reactions are uncontrolled
  - DEATH
- If body temp too low:
  - Cellular reactions too slow
  - DEATH



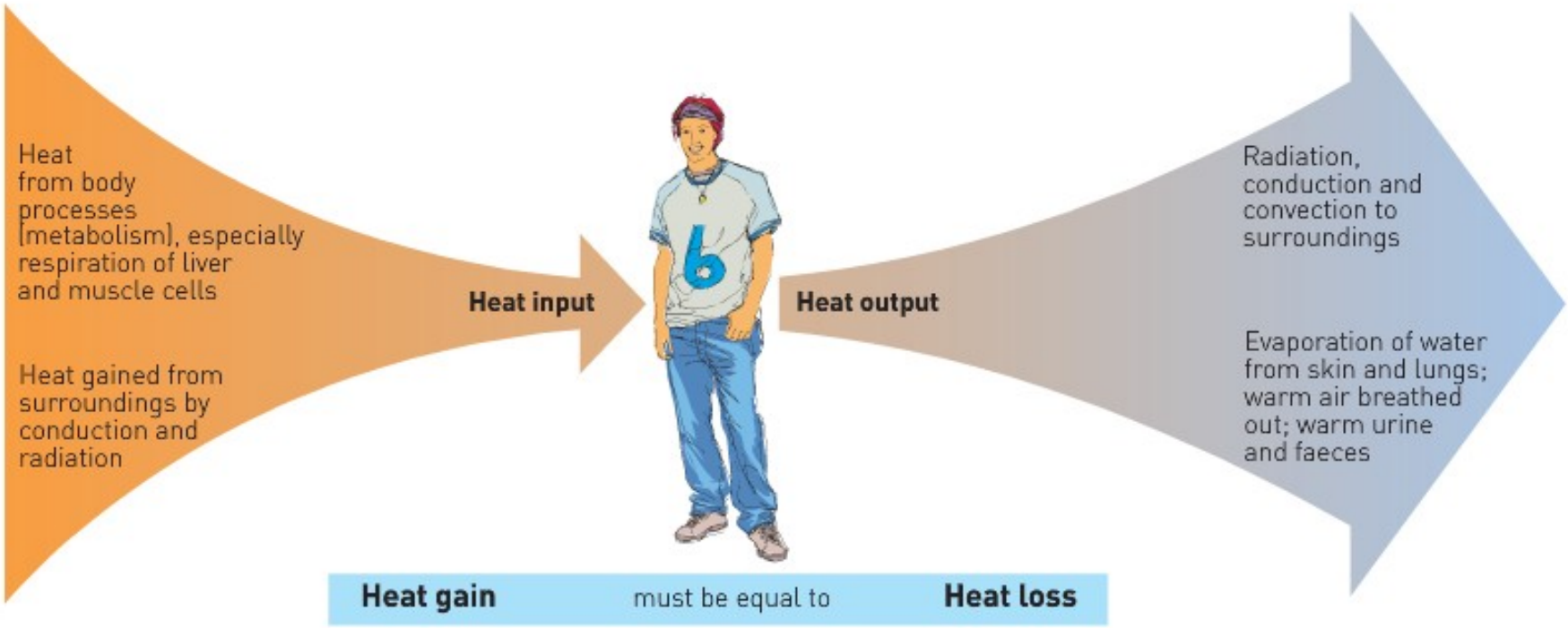


Figure 7.2 A constant human body temperature is achieved by a balance between heat gain and heat loss.

Heat Production

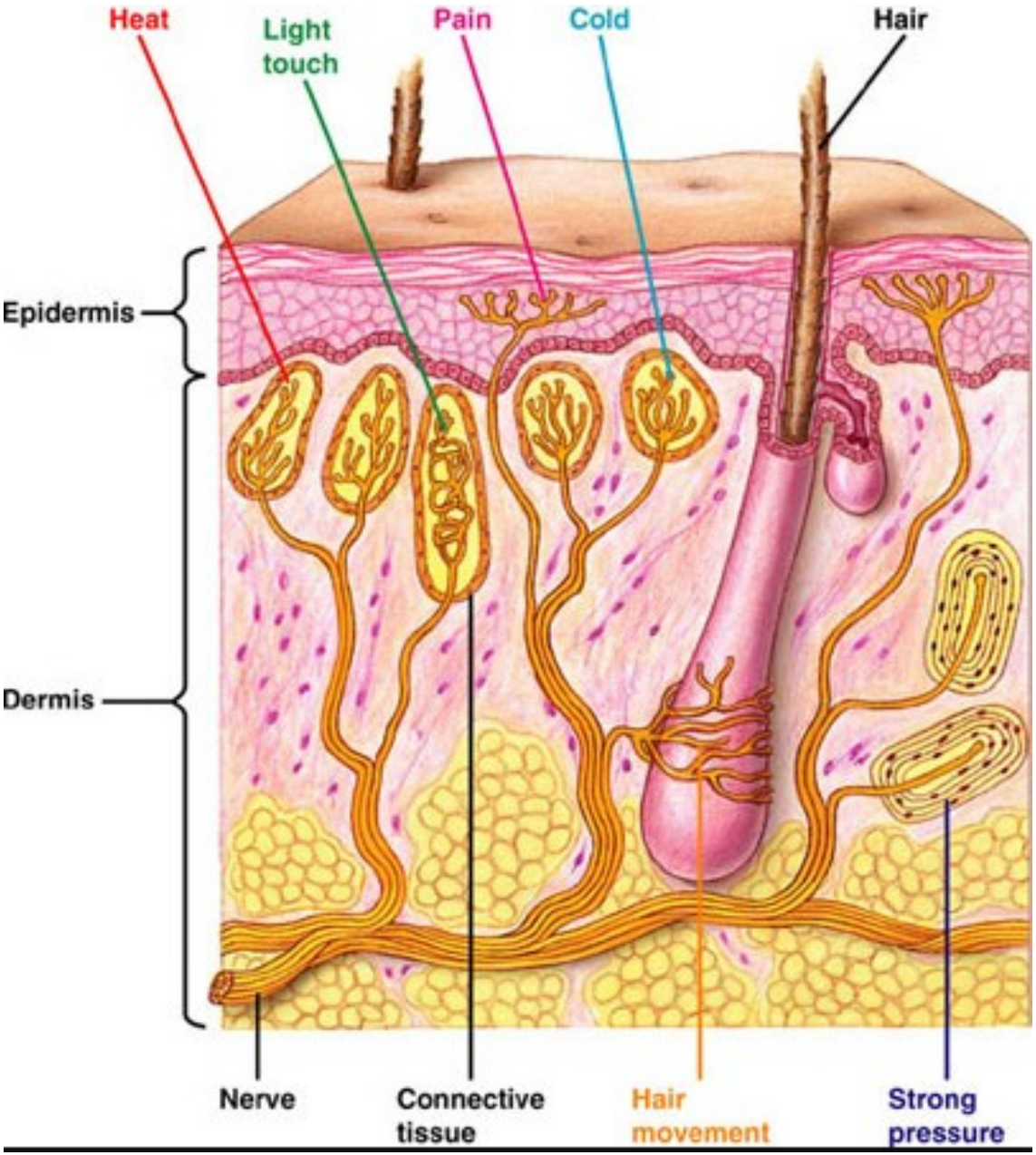
- During cellular respiration (metabolism)
- When metabolic rate increases, heat production increases.
  - Eg during Exercise:
    - Muscles need energy to contract
    - Cells respire more
    - More heat produced
    - Body heat rises



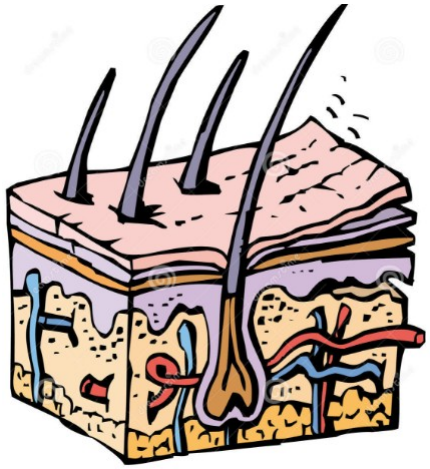
  - Things that can increase metabolic rate and body heat include:
    - Exercise
    - Stress
    - Sympathetic NS stimulation

Temperature receptors (thermoreceptors)

- Peripheral thermoreceptors:
  - In skin and other surface tissues
  - Detect temperature of external environment
    - Cold receptors
    - Heat receptors
  - Send signals to hypothalamus, which then acts to regulate body temperature
- Central thermoreceptors
  - In hypothalamus, spinal cord, abdominal organs
  - Send signals to hypothalamus, which then acts to regulate body temperature


Heat
Light touch
Pain
Cold
Hair
Epidermis
Dermis
Nerve
Connective tissue
Hair movement
Strong pressure

Skin and temperature regulation

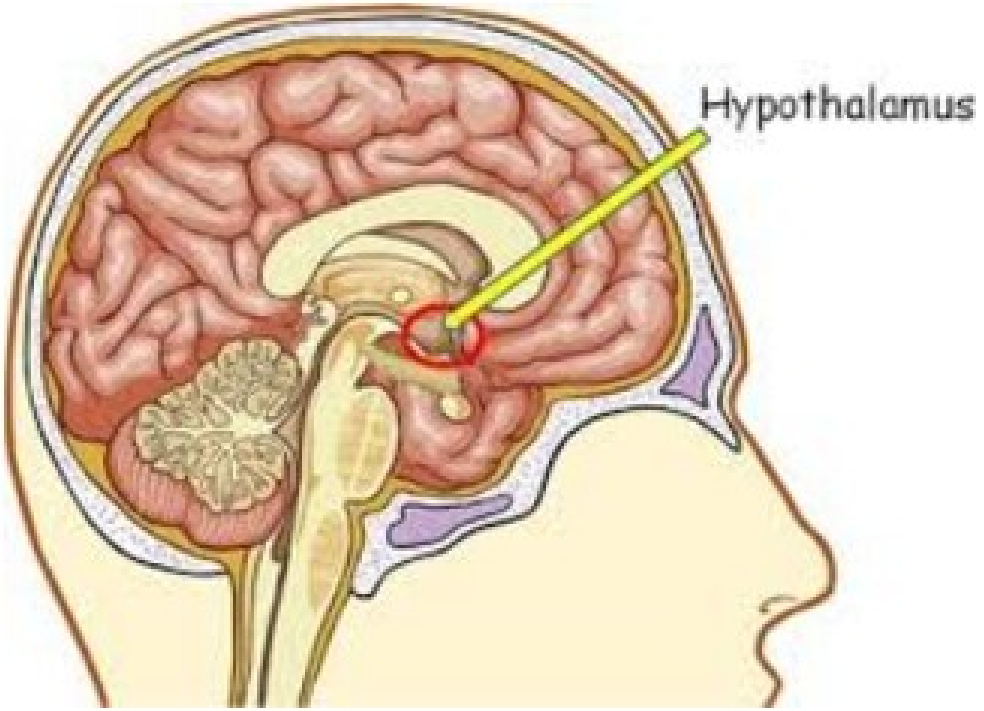


- Large surface area for heat exchange
  - Conduction
  - Convection
  - Radiation
- Blood vessels help exchange heat between external and internal environments
  - Diameter of blood vessels can be controlled to increase or decrease this exchange
- Sweat glands in skin produce sweat
  - Evaporative cooling!

Note: evaporation from lungs also causes heat loss

Preventing body temp from falling

- thermoreceptors in skin send message that skin is cold to **hypothalamus** via sensory nerves



Hypothalamic responses to falling temperature

- Vasoconstriction
  - Sympathetic stimulation from hypothalamus causes blood vessels to constrict
  - Decreases flow of warm blood to skin, so less heat exchange

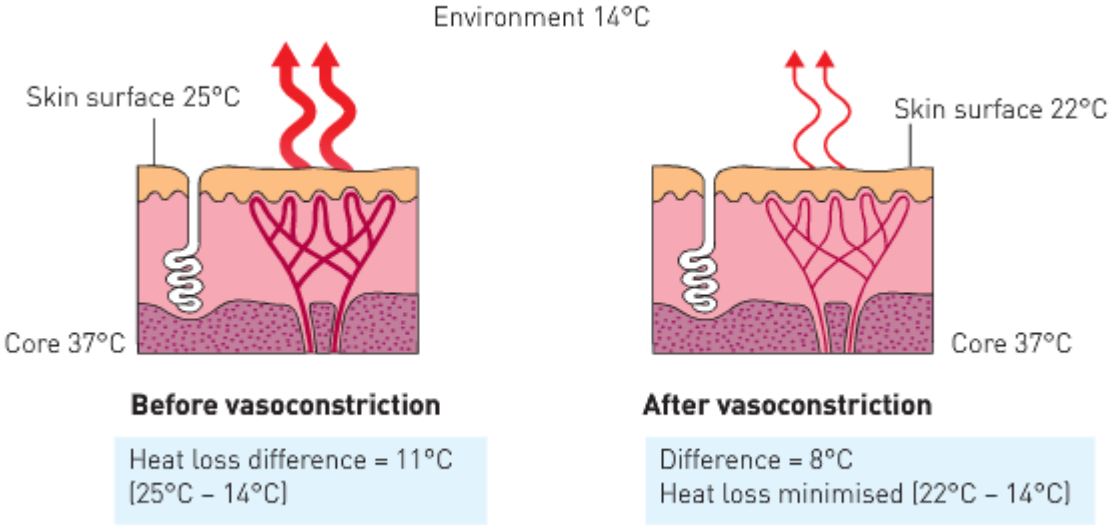
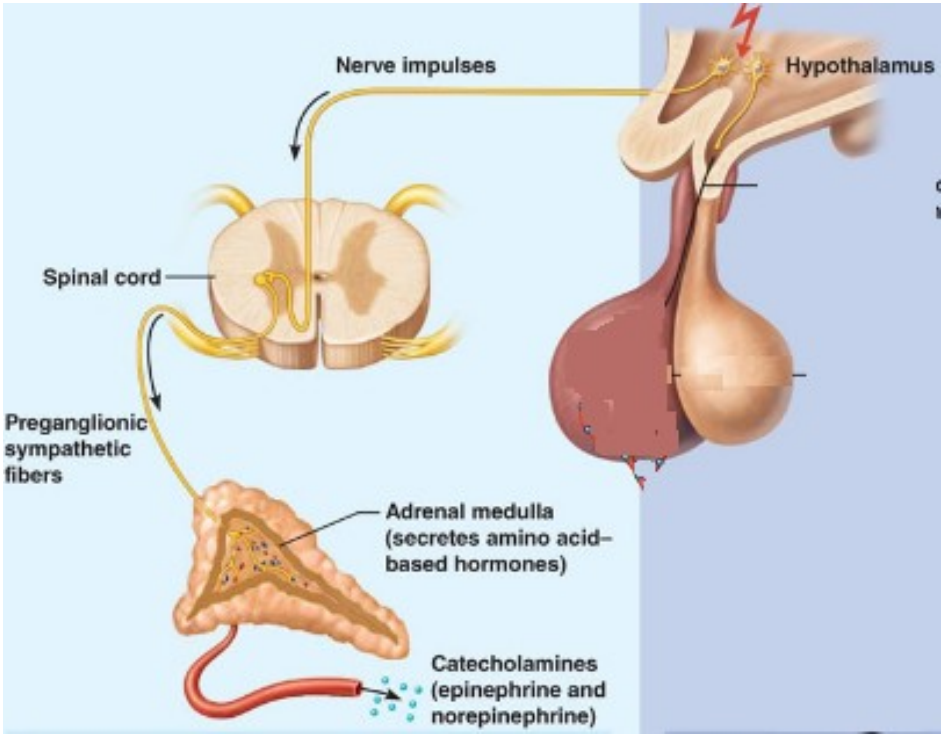


Figure 7.4 The effect of vasoconstriction in the skin

Hypothalamic responses to falling temperature

- Sympathetic NS stimulation of adrenal medulla
  - Adrenalin and Noradrenalin release
  - Increase in cellular metabolism and heat production
  - Counteracts heat loss



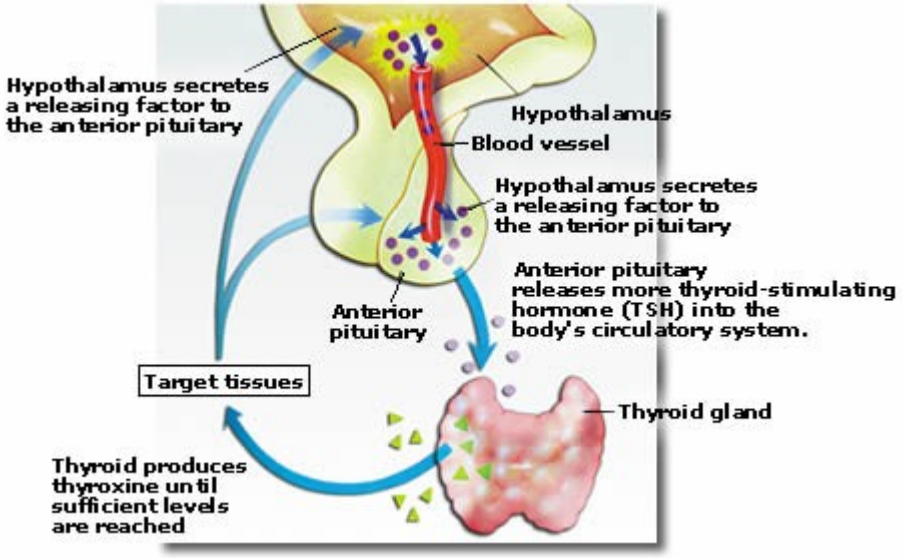
Hypothalamic responses to falling temperature

- Shivering
  - Hypothalamus sends message to brain centres that control skeletal muscle tone
  - Results in shivering – quick, repeated contraction of muscles
  - Heat is released from muscles
  - Can be influenced by conscious thought from cerebral cortex



Hypothalamic responses to falling temperature

- Increase in thyroxine production
  - Hypothalamus sends message to anterior pituitary
  - TSH (Thyroid Stimulating Hormone) released
  - Stimulates Thyroid
  - Thyroxine released
  - Increases metabolic rate and therefore heat.



Behavioural responses to falling temperature

- Conscious awareness of feeling cold leads to behavioural changes to minimise heat loss:
  - Seeking shelter from wind/rain/cold weather
  - Putting on clothing to insulate against heat loss
  - Curling up to decrease surface area for heat loss

Homeostasis when body temperature falls

Stimulus

Falling body temperature

Receptor

Detected by thermoreceptors in skin and hypothalamus

Message

Sent from skin via sensory NS to hypothalamus

Hypothalamus also detects blood temp.

Modulator (Control Centre)

Hypothalamus receives message that body temp is falling, and activates warming responses

Message

Sent mostly via Autonomic NS

Signals also sent to cerebral cortex - conscious awareness of temp.

Effector

Cerebral Cortex
Endocrine glands eg Thyroid
Blood Vessels in skin

Response

Body temp rises due to :
Conscious responses eg put on jumper, seek shelter, huddle up
Increased metabolic rate via Thyroxine release
Blood vessels in skin constrict - retain heat
Shivering - rapid muscle movements burn energy to create heat.

Feedback

Negative feedback. Body temp returns to normal, stimulus no longer present, cycle stops



Homeostasis when body temperature falls



Stimulus — Falling Body Temperature

Receptor — Thermoreceptors in Skin (surface) and Hypothalamus, Spinal Cord and Abdominal Organs (core)

Message — Sensory Nervous System

Control Centre / Modulator — Hypothalamus; Cerebral Cortex Conscious Awareness)

Message — Sympathetic Nervous System; Neurons within brain; TrH to ant.pit, then TSH; Somatic Motor NS

Effector — Blood Vessels; Adrenal Medulla; Brain centres for muscle tone; Thyroid; Skeletal Muscles

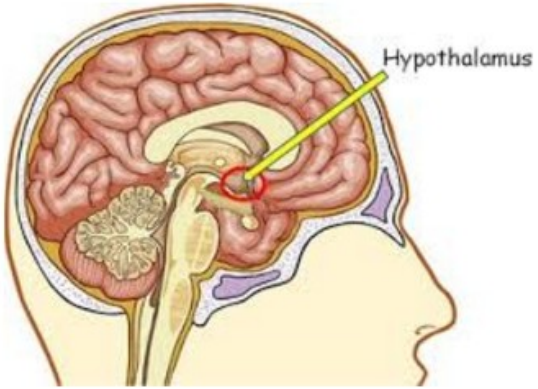
Response

- Vasoconstriction decreases blood flow to skin, reducing heat exchange
- Adrenalin and Nor-adrenalin release leads to increased cellular metabolism and heat production to counter-act heat loss
- Shivering leads to re-lease of heat from mus-cles to increase body temp
- Produces thyroid hormones that increase metabolic rate and therefore heat produc-tion.
- Behavioural responses such as seeking shelter, huddling, putting on clothes, turning on heater reduce heat loss.

Feedback — Negative Feedback: Body temperature returns to normal, stimulus no longer present, process ends.

Preventing body temp from rising

- Heat receptors in skin send message to **hypothalamus** via sensory nerves



- Most heat is lost through the skin, some lost via lungs and urine/faeces

Hypothalamic responses to rising temperature

- Vasodilation
  - Blood vessels in skin dilate
  - More blood can move through
  - Allows heat to be lost to surrounding environment

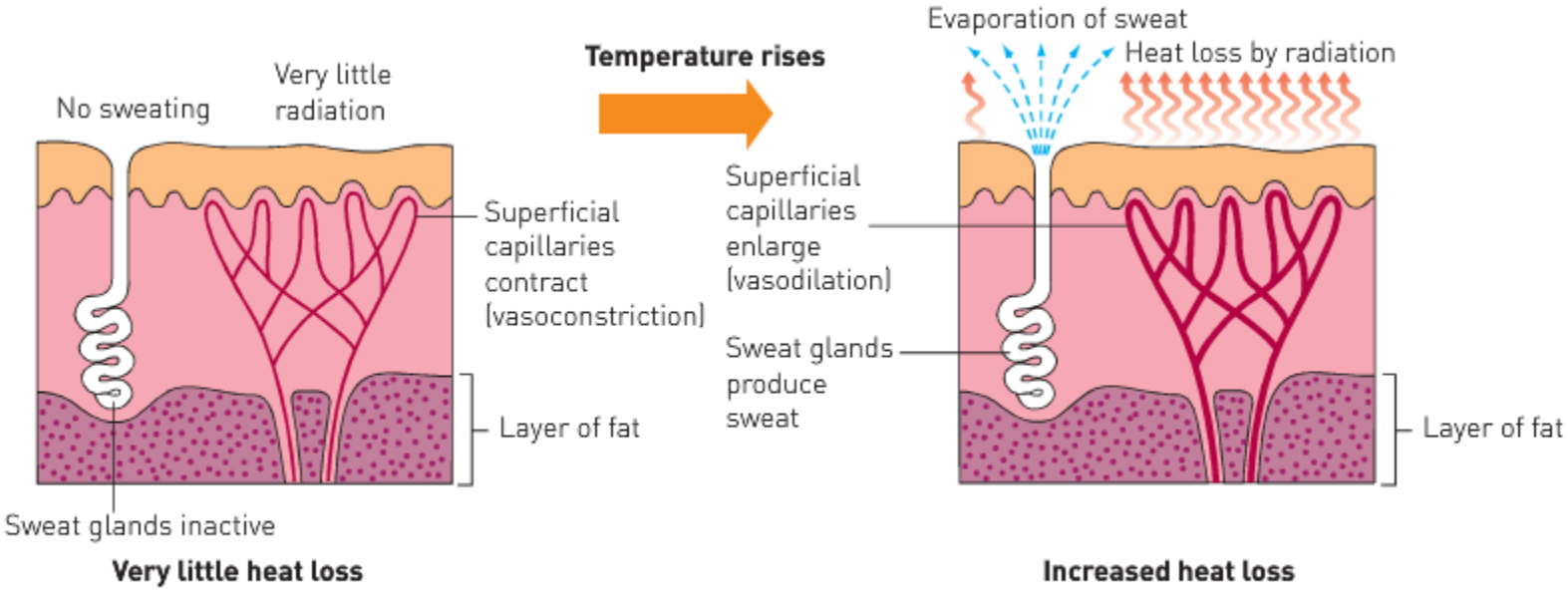


Figure 7.5 Heat loss from the skin can be increased by vasodilation of skin blood vessels and by sweating.

Hypothalamic responses to rising temperature

- Sweating
  - Needed for environmental temps above 28°C
  - Only option for cooling if environ above 37°C
  - Evaporation of liquid from skin accelerates heat transfer from body to environment
  - Works better in low humidity
- Longer term…
  - Decrease in metabolic rate due to reduced thyroxine

Behavioural responses to rising temperature

- Conscious awareness of feeling too hot leads to behavioural changes to allow cooling:
  - Wearing fewer clothing layers
  - Seeking cooler environments
  - Reducing physical activity

Homeostasis when Body Temp Rises

Stimulus

Rising body temperature

Receptor

Thermoreceptors in skin and hypothalamus

Message

Sent from skin and internal organs to hypothalamus via sensory NS. Hypothalamus also detects temp of blood.

Modulator (Control Centre)

Hypothalamus receives message that body temp is reising and activates cooling responses

Message

Sent via autonomic NS

Effector

Cerebral Cortex (conscious awareness)
Endocrine glands eg Thyroid
Blood Vessels in skin
Sweat Glands

Response

Body temp falls due to:
Conscious removal of clothing, shade seeking

Blood vessel dilation in skin - heat exchange
Sweat gland activity - evaporative cooling

Over time, metabolic rate may decrease - reduced Thyroxine

Feedback

Negative Feedback
Body temp returns to normal, stimulus ceases, cycle stops



Homeostasis when body temperature rises



Stimulus

Rising Body Temperature

Receptor

Thermoreceptors in Skin (surface) and Hypothalamus, Spinal Cord and Abdominal Organs (core)

Message

Sensory Nervous System

Control Centre / Modulator

Hypothalamus

Cerebral Cortex Conscious Awareness)

Message

Autonomic Nervous System

Negative feedback

Somatic Motor NS

Effector

Blood Vessels

Sweat Glands

Thyroid

Skeletal Muscles

Response

Vasodilation allows greater blood flow to skin and heat exchange via convection and radiation

Sweat secretion allows evaporative cooling of skin surface.

Reduction in thyroid hormone production decreases metabolic rate and therefore heat production (longer term response)

Behavioural responses such as wearing fewer clothing layers, seeking shade/cool environment, reducing physical activity.

Feedback

Negative Feedback: Body temperature returns to normal, stimulus no longer present, process ends.

In cold conditions

To decrease heat loss

- Constriction of blood vessels in the skin
- Reduction in sweating
- Conscious behaviour such as putting on a jumper, sheltering from wind
- Reduction of surface area such as by curling into a ball



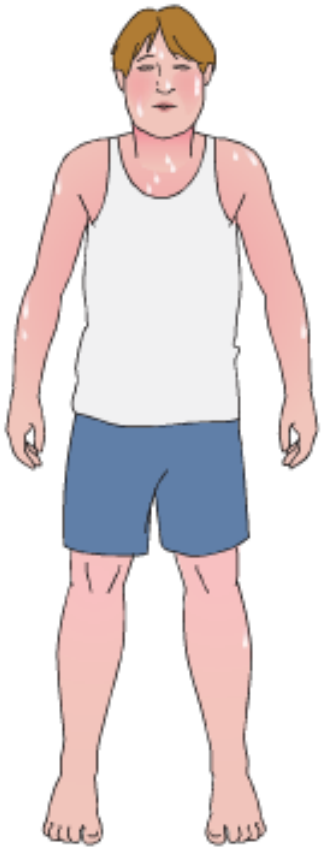
To increase heat production

- Shivering
- Increase in voluntary activity
- Increased metabolic rate (long-term response)

In hot conditions

To increase heat loss

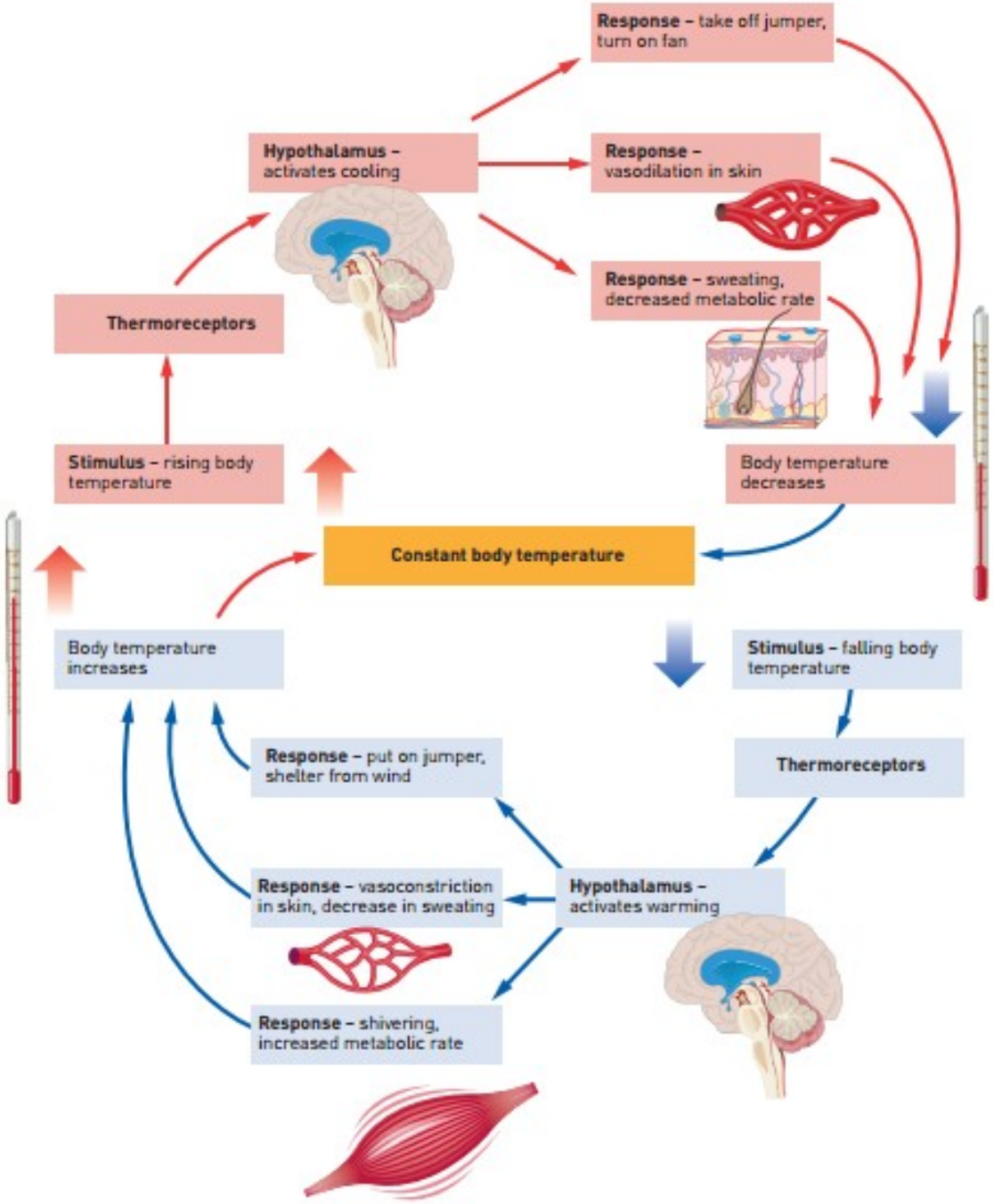
- Dilation of blood vessels in the skin
- Sweating
- Conscious behaviour such as removing clothing, turning on a fan
- Increase surface area by spreading out



To decrease heat production

- Decrease in voluntary activity
- Decreased metabolic rate (long-term response)

Figure 7.6 Mechanisms for regulating temperature


Response – take off jumper, turn on fan
Hypothalamus – activates cooling
Response – vasodilation in skin
Response – sweating, decreased metabolic rate
Thermoreceptors
Stimulus – rising body temperature
Body temperature decreases
Constant body temperature
Body temperature increases
Stimulus – falling body temperature
Response – put on jumper, shelter from wind
Thermoreceptors
Response – vasoconstriction in skin, decrease in sweating
Hypothalamus – activates warming
Response – shivering, increased metabolic rate

Temperature Tolerance

- Core body temp above 42°C is dangerous and will result in death if prolonged.
- Heat stroke:
  - when the body is unable to manage temperature due to extreme external conditions
  - Treatment – careful cooling
- Heat exhaustion:
  - Less severe
  - Dehydration from sweating → collapse
- Hypothermia:
  - Temps below 32°C.
  - Death if prolonged